

End-to-End Vendor Performance and Inventory Analytics: A Comprehensive Case Study

1. Executive Summary

Effective inventory management and vendor performance evaluation are critical determinants of profitability in the retail and wholesale sectors. This project outlines an end-to-end data analytics pipeline designed to process, clean, and analyze a massive volume of procurement and sales data. By employing advanced SQL querying (ETL processes) and Python-based statistical analysis, this study identifies critical inefficiencies in capital allocation, highlights high-margin promotional opportunities, and validates pricing strategies, ultimately delivering actionable insights to optimize supply chain profitability.

2. Data Overview and Processing

The analysis is built upon a substantial dataset comprising millions of transactional records across sales, purchases, and vendor invoices.

- **Data Volume and Transformation:** The raw data was ingested into a local SQLite database using memory-efficient chunking techniques. Through optimized SQL queries utilizing Common Table Expressions (CTEs), the transactional data was aggregated into a highly condensed master summary table consisting of 10,692 unique records and 16 features.
- **Data Cleaning:** To ensure statistical validity, anomalous records—such as products with zero sales volume or negative gross profit margins—were filtered out. The refined analytical dataset contains 8,565 high-quality records.
- **Descriptive Statistics (Filtered Dataset):**
 - **Gross Profit:** The mean gross profit per product stands at \$15,653.19, with a maximum reaching \$1,290,668.
 - **Profit Margin:** The average profit margin is highly robust at 38.72%, peaking at 99.71%.
 - **Stock Turnover:** The mean stock turnover ratio is 2.03. However, the presence of extreme outliers (up to 274.5) indicates the sale of historically accumulated inventory over the current reporting period.

3. Key Business Insights and Analytical Findings

A. The Impact of Bulk Purchasing on Unit Cost

An analysis of procurement volumes was conducted to evaluate the efficacy of bulk purchasing strategies. The data was segmented into three order size tiers (Small, Medium, Large) based on quantiles. The results demonstrated a drastic reduction in unit costs as order volumes increased:

- Small Orders: \$39.05 average unit price.
- Medium Orders: \$15.48 average unit price.
- Large Orders: \$10.77 average unit price.
- **Conclusion:** Bulk procurement effectively reduces the unit purchase price by approximately 72.4% compared to small-tier orders, thoroughly validating the company's volume discount strategies.

B. Capital Efficiency and Unsold Inventory

A critical assessment of inventory turnover ratios (specifically focusing on ratios < 1.0) revealed severe capital inefficiencies.

- **Tied Capital:** The total locked capital trapped in unsold inventory amounts to a staggering **\$13,993,308.64**.
- **Top Contributors:** A concentrated group of vendors is responsible for the majority of this tied capital. The top 10 vendors contributing to unsold inventory are:
 1. DIAGEO NORTH AMERICA INC (\$1,922,988.64)
 2. MARTIGNETTI COMPANIES (\$1,344,905.78)
 3. JIM BEAM BRANDS COMPANY (\$1,157,917.44)
 4. PERNOD RICARD USA (\$888,904.94)
 5. ULTRA BEVERAGE COMPANY LLP (\$772,078.78)
 6. E & J GALLO WINERY (\$635,224.69)
 7. WILLIAM GRANT & SONS INC (\$632,051.92)
 8. CONSTELLATION BRANDS INC (\$560,083.07)
 9. PERFECTA WINES (\$542,845.17)
 10. BROWN-FORMAN CORP (\$469,267.64)

C. Identification of High-Potential "Target Brands"

To uncover hidden revenue opportunities, the dataset was analyzed to find products exhibiting low sales volume alongside exceptionally high profit margins.

- **Thresholds applied:** Sales below the 15th percentile ($< \$562.67$) and profit margins above the 85th percentile ($> 63.41\%$).
- **Conclusion:** The analysis identified **233 specific brands** that meet these criteria. These products represent highly lucrative, untapped revenue streams that require immediate strategic marketing and promotional adjustments to stimulate sales volume.

D. Statistical Analysis of Profit Margins (Hypothesis Testing)

To determine if sales volume dictates profitability, vendors were segmented into Top Performers (top 25% by sales revenue) and Low Performers (bottom 25% by sales revenue). A 95% Confidence Interval (CI) was calculated for both groups:

- **Top Vendors:** Mean profit margin = 31.17% (95% CI: 30.74% to 31.61%).
- **Low Vendors:** Mean profit margin = 41.55% (95% CI: 40.48% to 42.62%).
- **Conclusion:** There is a statistically significant difference in profit margins. Vendors with lower overall sales volumes consistently maintain significantly higher profit margins. This indicates that low-volume vendors likely specialize in premium, niche products, whereas top-tier vendors rely on high-volume, lower-margin bulk sales.

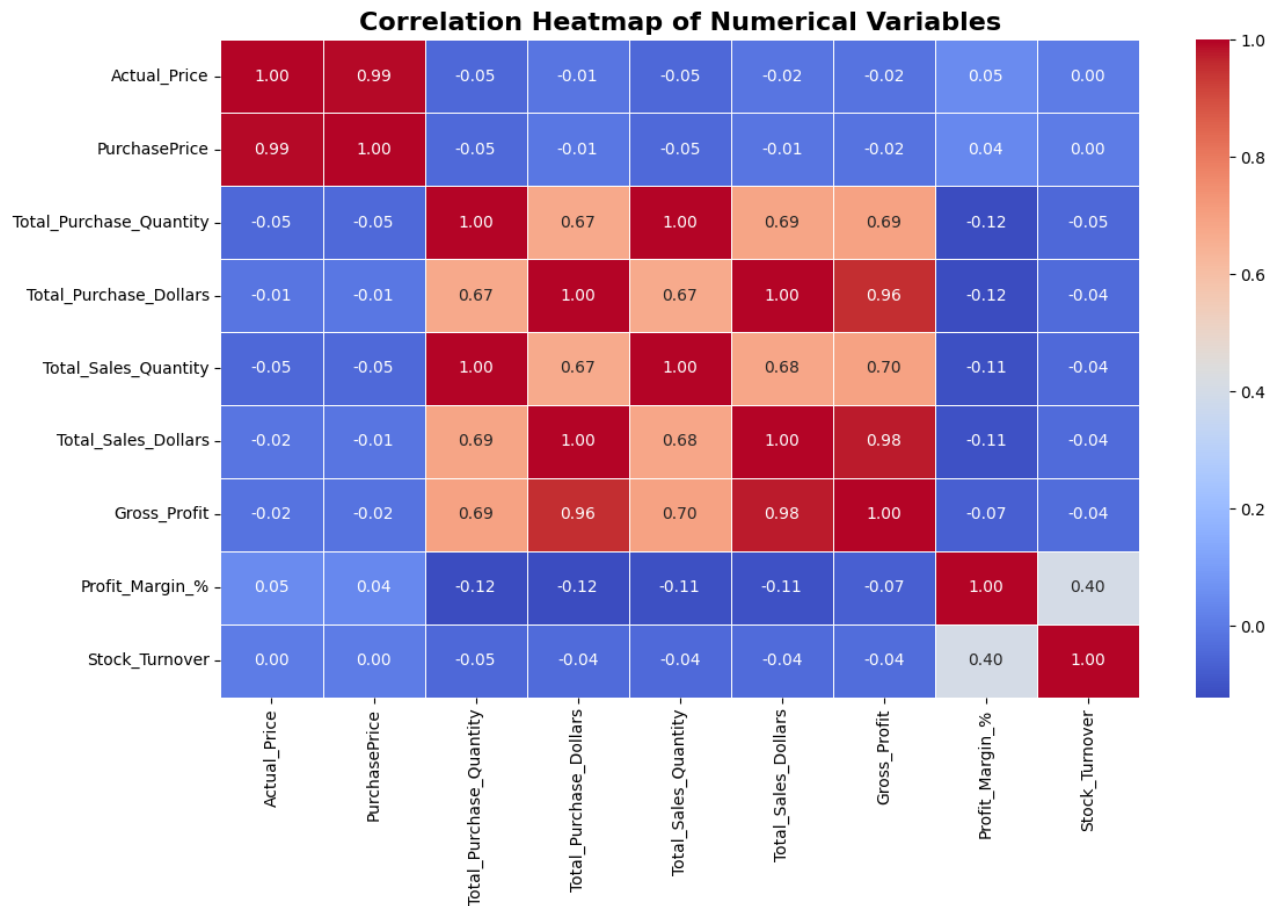
4. Strategic Recommendations

Based on the data-driven insights gathered, the following strategic actions are recommended:

1. **Inventory Liquidation:** Initiate immediate discussions with the top 10 vendors holding the \$13.9M in unsold inventory. Implement discount strategies, product bundling, or return-to-vendor negotiations to liquidate stagnant stock and free up working capital.

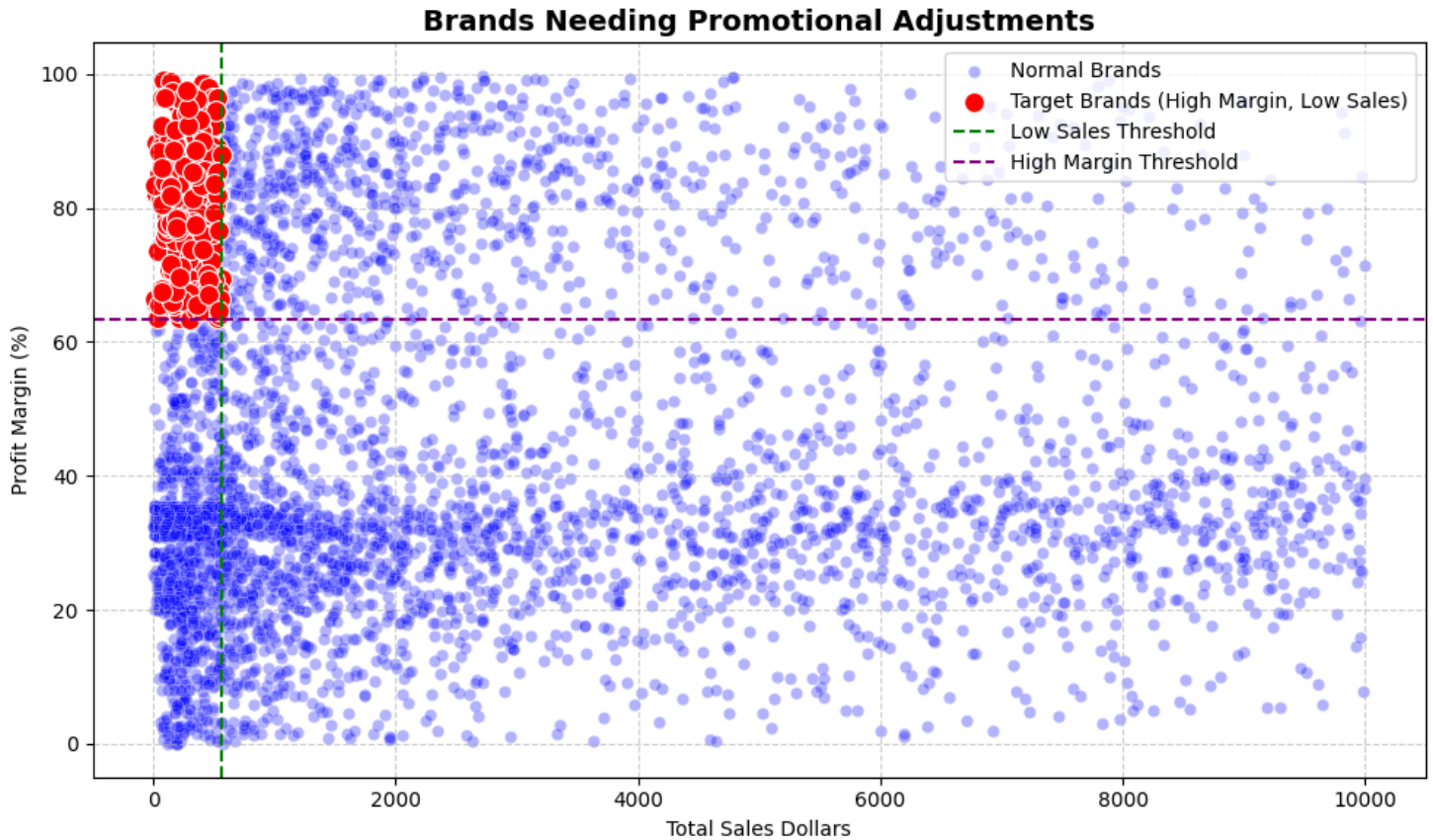
- 2. **Targeted Marketing Campaigns:** Allocate marketing budget to promote the 233 identified "high-margin, low-sales" brands. Minor price reductions or targeted advertising for these specific items will yield exponential returns.
- 3. **Procurement Optimization:** Continue to leverage large-volume purchasing to maintain the \$10.77 average unit cost, but strictly enforce turnover rate thresholds to prevent future capital lock-up.

1. Correlation Matrix of Key Performance Metrics



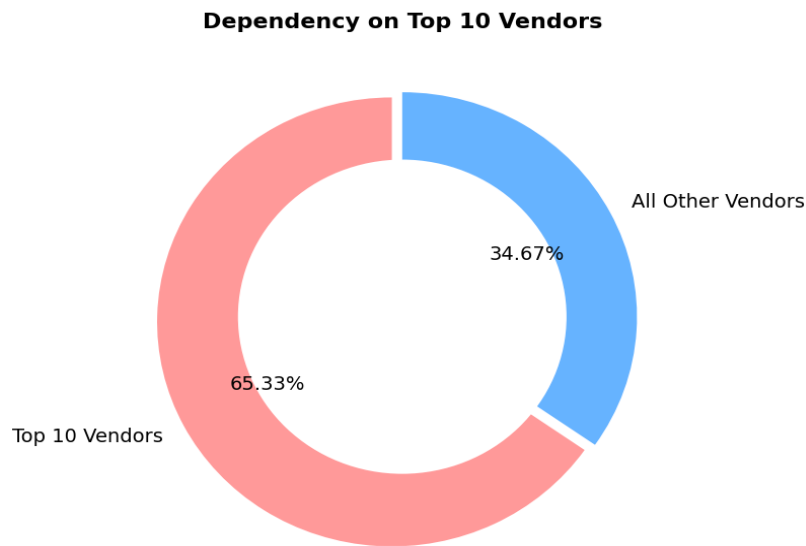
This heatmap illustrates the Pearson correlation coefficients between critical numerical variables within the dataset. A key observation is the strong positive correlation between purchase quantities and sales quantities, indicating generally efficient inventory movement. Conversely, there is a weak negative correlation between the unit sales price and the profit margin, suggesting that higher-priced items do not necessarily yield higher percentage margins, likely due to competitive pricing pressures and market elasticity.

2. Identification of Target Brands for Promotional Strategies

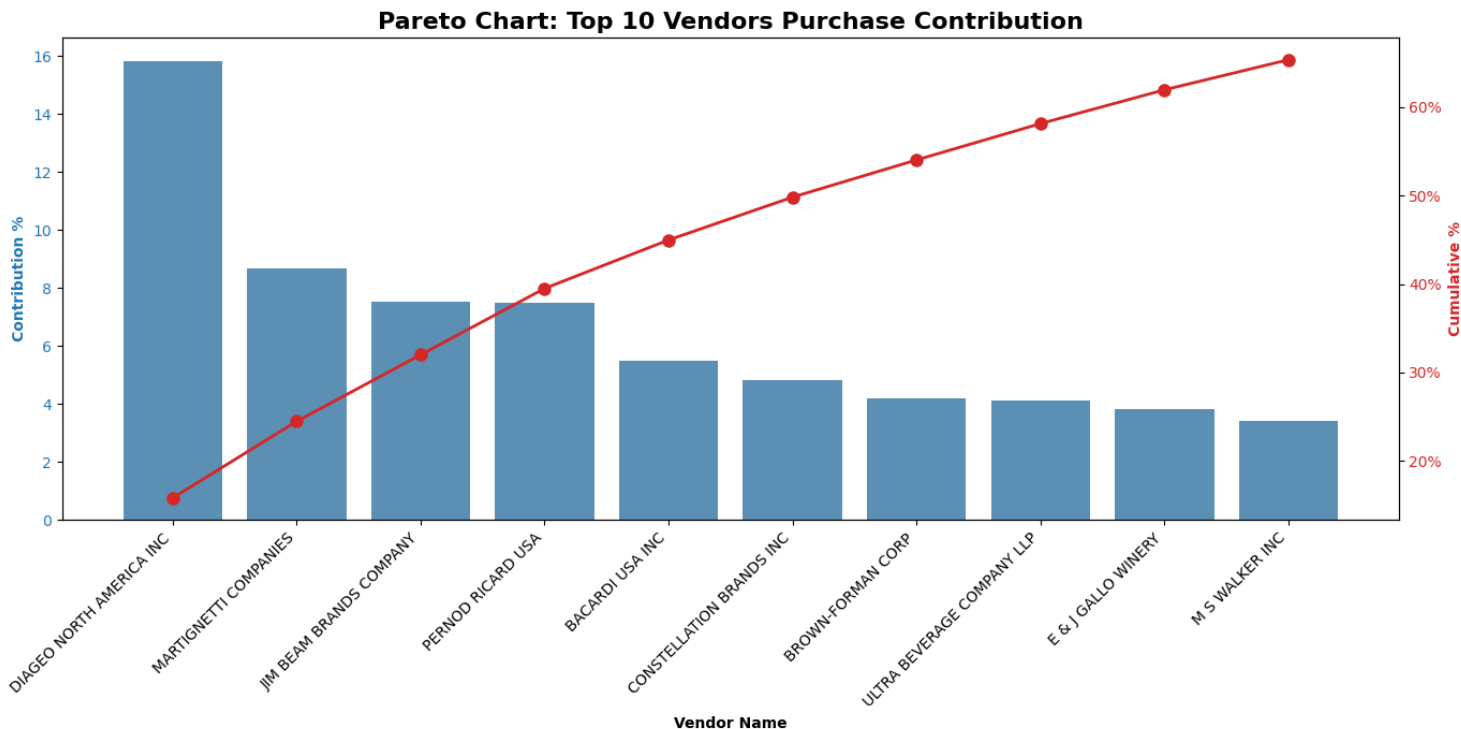


This scatter plot categorizes product brands based on total sales revenue and average profit margin. The highlighted data points represent the identified "Target Brands," which fall below the 15th percentile in sales volume but exceed the 85th percentile in profit margin. These specific products represent untapped revenue potential and are prime candidates for targeted marketing campaigns, bundling, or strategic price adjustments to drive sales volume without sacrificing profitability.

3. Pareto Analysis of Vendor Purchase Contribution



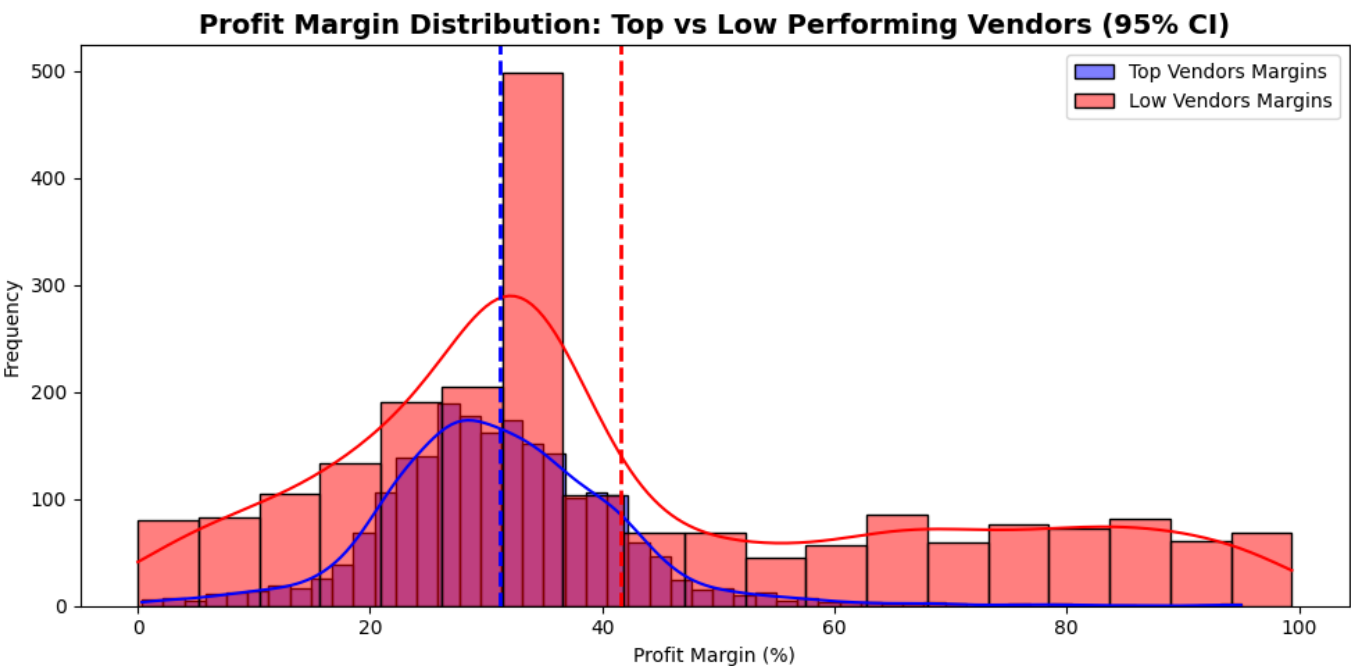
This combination bar and line chart applies the Pareto principle to vendor procurement. The bars represent the individual purchase contribution of the top ten vendors, while the superimposed line tracks the cumulative percentage of total procurement. The analysis reveals a significant supply chain dependency risk, demonstrating that approximately 66% of the company's total procurement budget is concentrated among a mere ten vendors out of the entire supplier base.



4. Procurement Dependency Distribution

This chart provides a macroscopic view of vendor dependency and procurement allocation. It visually contrasts the heavy reliance on the top ten vendors—who account for nearly two-thirds of the total purchase volume—against the remaining supplier pool. This unequal distribution underscores an operational vulnerability and highlights the strategic necessity for supply chain diversification to mitigate potential vendor-related disruptions.

5. Statistical Analysis of Profit Margins by Vendor Performance



This histogram displays the distribution of profit margins, comparing top-performing vendors (the top 25% by sales revenue) against low-performing vendors (the bottom 25% by sales revenue). The vertical lines indicate the 95% confidence intervals for the mean of each respective group. The statistical testing confirms a significant difference: low-volume vendors maintain substantially higher profit margins (averaging 41.55%) compared to top-tier vendors (31.17%). This divergence suggests distinct underlying business models, such as premium pricing for niche products versus volume-based discounting.

6. Predictive Modeling: Demand Forecasting and Feature Importance

These visualizations present the results of the Random Forest Regressor model developed for daily sales demand forecasting. The left panel details the feature importance, identifying the primary variables that drive daily sales, most notably historical sales data and average pricing. The right panel illustrates the model's performance by comparing predicted sales quantities against actual historical data. The tight clustering along the trend line demonstrates the model's predictive accuracy, confirming its viability as an automated tool for optimizing future inventory replenishment and minimizing tied capital.